Task 1: DNA Sequence Analysis

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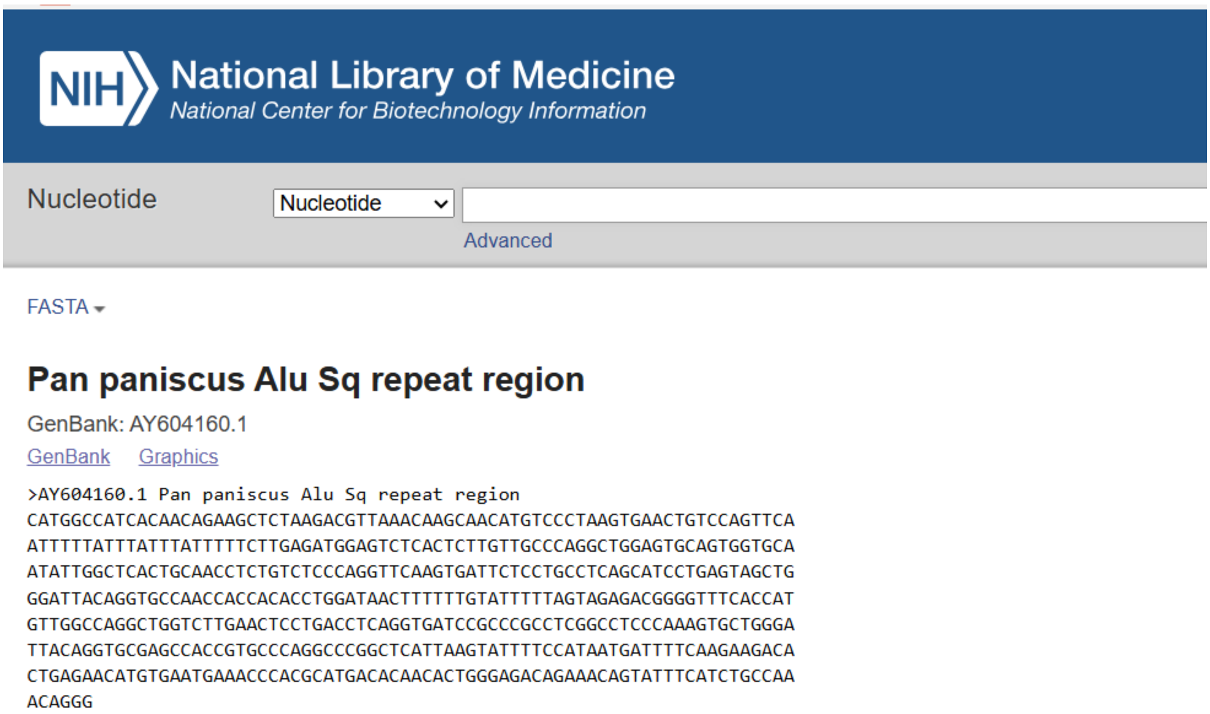
Domain: Bioinformatics

Step 1. Retrieve the sequence

Search the accession **AY604160.1** in NCBI Nucleotide using:

Enter **AY604160.1** in the search box to open the sequence record. NCBI nucleotide records can be retrieved directly using accession numbers.



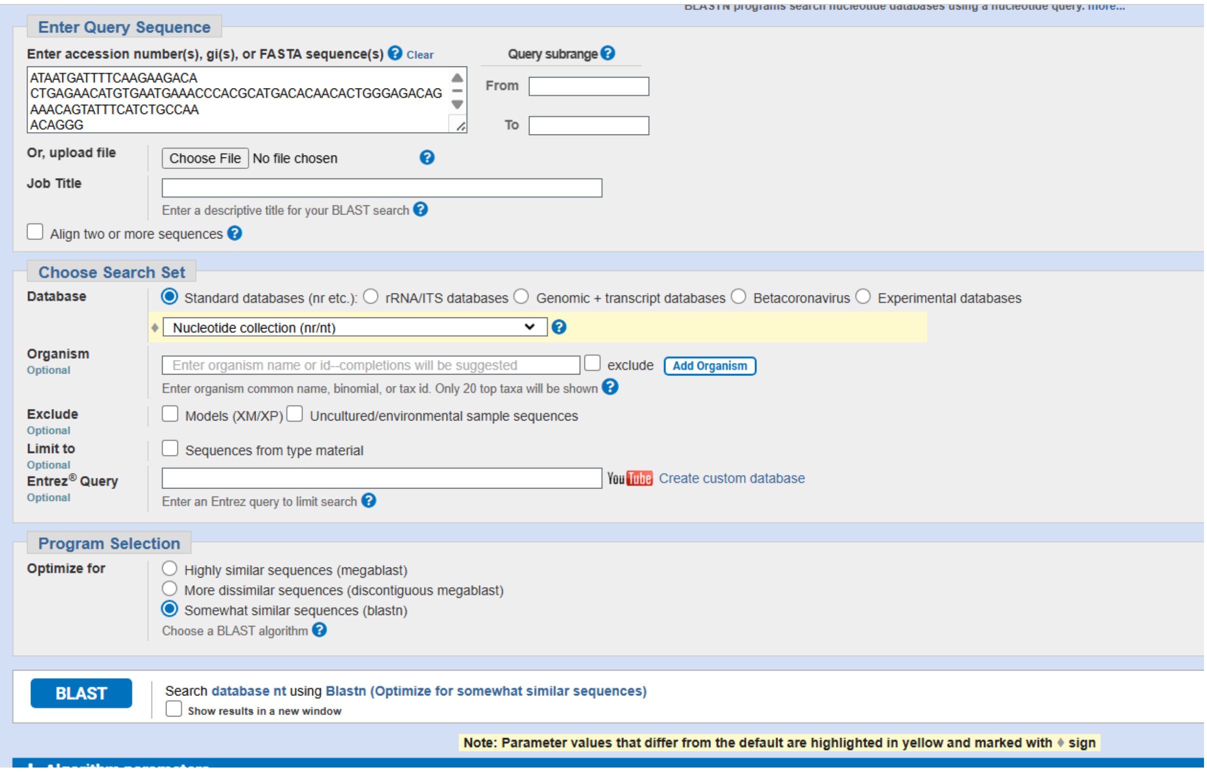
Step 2. Download the FASTA sequence

1. Open the record.
2. Click **Send to** → **File**.
3. Select **FASTA**.
4. Save the file.

Step 3. Run BLAST

Go to: NCBI BLAST

Choose: **Nucleotide BLAST (BLASTn)**
Paste the FASTA sequence.
Keep the default settings.
Click **BLAST**.



Step 4. Record these values

From the first (best) alignment note:

Parameter	Description
Query Coverage	100%
Percent Identity	100%
E-value	0.0
Max Score	954

NIH National Library of Medicine
National Center for Biotechnology Information

Log i

BLAST ® » blastn suite » results for RID-3Z0FAEMR016

Home Recent Results Saved Strategies H

Edit Search Save Search Search Summary

How to read this report? BLAST Help Videos Back to Traditional Results P

Job Title	**Nucleotide Sequence**
RID	3Z0FAEMR016 *Search expires on 06-28 15:55 pm* Download All
Program	BLASTN Citation
Database	nt See details
Query ID	lcl|Query_6252891
Description	None
Molecule type	dna
Query Length	496
Other reports	Distance tree of results MSA viewer

Filter Results

Organism *only top 20 will appear* exclu

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments

Download Select columns Show 10

select all *10 sequences selected*

GenBank Graphics Distance tree of results MSA View

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accessio
Homo sapiens Alu repeat region	Homo sapiens	954	954	100%	0.0	100.00%	496	AY604158.
Gorilla gorilla Alu repeat region	Gorilla gorilla	931	931	100%	0.0	99.19%	497	AY604157.
Homo sapiens chromosome 8 clone CH17-241F1, complete sequence	Homo sapiens	896	9738	100%	0.0	97.98%	243888	AC270296
Homo sapiens chromosome 8, clone RP11-212F11, complete sequence	Homo sapiens	896	4599	100%	0.0	97.98%	167547	AC112673.
Pan paniscus Alu Sq repeat region	Pan paniscus	539	539	62%	1e-147	97.07%	307	AY604164.
Eukaryotic synthetic construct chromosome 13	eukaryotic synthetic construct	421	1.067e+05	64%	2e-112	92.96%	96089878	CP034516
Eukaryotic synthetic construct chromosome 13	eukaryotic synthetic construct	421	1.067e+05	64%	2e-112	92.96%	96089878	CP034491
Homo sapiens chromosome 2 clone CH17-218L2, complete sequence	Homo sapiens	421	21183	63%	2e-112	92.68%	233881	AC277931
Human DNA sequence from clone RP11-169O17 on chromosome 13, complete seq...	Homo sapiens	421	8190	62%	2e-112	92.96%	190765	AL359763.
Homo sapiens chromosome 5 clone CH17-458O9, complete sequence	Homo sapiens	417	35059	62%	3e-111	91.58%	209864	AC277898